BRIEF ON NASARAWA STATE GOVERNMENT DATA CENTER & FIBER OPTIC NETWORK PROJECT

CYNOX IT LTD is proposing to the NASARAWA State Government the establishment of a state-owned Fiber Optics Network Infrastructure and data center. The fiber optic cables will stretch across major roads and streets in the metropolis, with the data center serving as the termination point.

FIBER OPTIC NETWORK INFRASTRUCTURE

- At the heart of the fiber optic network lies a state-of-the-art 96-core cable, serving as the backbone for transmitting vast amounts of data.
- A double duct system will be utilized, which involves installing two parallel ducts. This configuration offers an additional layer of physical protection for the cables.
- Manholes/handholes serve as access points facilitating easy maintenance, repair, and expansion of the network.

NASARAWA STATE DATA CENTER

The proposed Data Center will be a cutting-edge facility established by the State Government. It will function as a centralized hub for managing, storing, and processing data, enabling seamless communication and improved service delivery across government departments.

- The Data Center will offer cost-effective and secure services including:
    - **Co-location Services:** The Data Center can provide co-location services, offering secure space for businesses and organizations across various sectors such as Aviation, health, agriculture, education, finance, and more, to house their data and IT equipment.
    - **Hosting Services**: The Data Center can offer hosting services, enabling businesses and organizations to host their websites, applications, and databases on the facility's robust infrastructure.
    - **Call Center Services:** The Data Center can provide call center services, offering businesses a centralized facility to manage customer support, telemarketing, or help desk operations.

Address: Plot 2978, Logone Close, Off Erie crescent, Off Nile Street, Maitama- Abuja.
Email:info@cynoxit.com.ng Website: www.cynoxit.com.ng

- **Surveillance Services**: The Data Center can offer surveillance services, utilizing advanced security systems, video monitoring, and access control mechanisms.
- **ISPs**: The Data Center can serve as an interconnection point for internet service providers (ISPs) in NASARAWA State.

AGRICULTURE: EXPLORING ICT INNOVATION IN FARMING

The data center can be instrumental in aiding farmers in NASARAWA State through various means:

- **Data Analytics for Precision Farming:** By collecting and analyzing data related to soil health, weather patterns, crop growth, and market trends, farmers can make informed decisions to optimize their yields and resource utilization.
- **Access to Agricultural Information:** A data center can host platforms or databases containing valuable agricultural information, such as best practices, crop varieties suitable for local conditions, pest and disease management strategies, and market prices. This information can be accessed by farmers even in remote areas, helping them improve their farming techniques and profitability.
- **Remote Monitoring and Control:** With the help of IoT devices and sensors deployed in fields, data centers can remotely monitor various parameters like soil moisture, temperature, and crop health. This real-time data can enable farmers to take timely actions such as irrigation scheduling, pest control, and nutrient management.
- **Market Access and Supply Chain Optimization:** By integrating data from various stakeholders in the agricultural supply chain, including farmers, distributors, and retailers, data centers can facilitate efficient logistics and market access for farmers. This can help reduce post-harvest losses and ensure fair prices for their produce.
- **Farm Management Software:** Software platforms and mobile apps help farmers with tasks such as crop planning, inventory management, financial tracking, and equipment maintenance scheduling.

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- **Climate Resilience and Risk Management:** Data centers can host predictive models and tools that analyze historical weather data and climate projections to help farmers anticipate and mitigate the impacts of climate change, such as droughts, floods, and extreme temperatures.
- **Financial Inclusion and Access to Credit:** By aggregating and analyzing data on farmers' productivity, income, and creditworthiness, data centers can support financial institutions in providing tailored financial products and services to farmers, thereby promoting financial inclusion and investment in agriculture.

Overall, leveraging the capabilities of data centers can empower farmers in Africa with the information and tools they need to improve their livelihoods, increase food security, and contribute to sustainable agricultural development.

REVENUE GENERATION

The proposed Fiber Optics Network Infrastructure and Data Center can offer various opportunities for revenue generation and attraction of investors. Here are some key areas where revenue can be generated:

- **Data Hosting and Warehousing**: Revenue can be generated through monthly or annual hosting fees, which are determined by the allocated server resources, storage capacity, network bandwidth, and additional value-added services such as data backup and disaster recovery.
- **Data Transit**: we can offer data transit services to businesses, internet service providers (ISPs), and government agencies, charging fees based on bandwidth usage and data transfer volumes.
- **ISPs**: Revenue can be generated through network interconnection fees, bandwidth utilization charges, and value-added services like IP transit or content delivery.
- **Data Center as a Service (DCaaS):** we can generate revenue by charging customers for resource utilization and service-level agreements.
- **Business Continuity and Disaster Recovery**: we can offer customized solutions based on customer requirements, charging fees for service subscriptions, data replication, and recovery processes.

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- **Co-location Services**: Customers can lease racks, cabinets, or private suites within the facility, paying regular fees for occupying the space and utilizing power and cooling resources.
- **Call Center Services**: Revenue can be generated through service contracts or billing based on the number of agents, call volume, or the duration of call center usage.
- **Surveillance Services**: Customers, such as businesses or government agencies, can lease the surveillance infrastructure and pay fees based on the number of cameras, storage requirements, and additional security features.
- **Conduit Lease**: Revenue can be generated through long-term leasing contracts and recurring fees based on the leased conduit infrastructure. Telecom operators pay for the right to utilize the conduits and install their fiber optic cables, allowing them to extend their network reach and offer broadband services to a wider customer base.
- **Dark Fiber Lease**: Revenue can be generated through leasing agreements, where telecom operators pay fees based on the length and capacity of the leased dark fibers. Dark fiber lease provides flexibility and scalability for telecom operators to tailor their network infrastructure according to their specific requirements, and it also enables them to offer differentiated services to their customers.
- **Fiber-to-the-Home (FTTH) Connectivity**: Revenue can be generated through subscription fees from homeowners, offering various internet packages based on speed and data limits.
- **Community Hotspots**: Revenue can be generated through advertising partnerships, sponsorship agreements, or charging fees for premium access or value-added services within the hotspot.

The outcome of the Data Center and Fiber Optics project will significantly support the efforts of the Federal Government towards achieving the goals of the digital economy and broadband penetration in rural areas, as outlined in various national policies and strategies. These include the National Digital Economy Policy and Strategy (NDEPS, 2020-2030), the National Broadband Plan (NBP 2020-2025), the NCC's Strategic Vision Plan (SVP2023-2025), NITDA's Strategic Roadmap and Action Plan (SRAPs 2021-2025), and the National Policy for the Promotion of Indigenous Content.

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